14.2 Phase Changes

Question Paper

Course	CIE A Level Physics
Section	14. Temperature
Topic	14.2 Phase Changes
Difficulty	Easy

Time allowed: 50

Score: /36

Percentage: /100

Question 1a

Specific heat capacity and latent heat capacity are described in Table 1.1.
Determine the correct definitions and equations by placing them in the correct boxes in Fig 1.1.

Table 1.1

$\Delta Q = mc\Delta\theta$	The thermal energy required to change the state of 1 kg of mass of a substance without any change of temperature.
$Q = Lm$	The amount of thermal energy required to raise the temperature of 1 kg of a substance by 1 °C.

Fig 1.1

	Equation	Definition
Specific Latent Heat Capacity		
Specific Heat Capacity		

[4 marks]

Question 1b

A student is trying to explain specific heat capacity to a friend.

Identify, by placing a tick (✓) next to the correct statements about specific heat capacity.

Statement	(✓) here if the statement is correct
All substances have the same specific heat capacity.	
The heavier the material, the more thermal energy is required to raise its temperature.	
The larger the change in temperature, the higher the thermal energy will be required to achieve this change.	
Specific heat capacity is mainly used in gases.	

[2 marks]

Question 1c

The student is confused about how to explain specific latent heat capacity.

Identify, by circling the best option, the explanations that correctly describe specific latent heat capacity.

(i)
When a substance changes state there is **a temperature change / no temperature change**

[1]

(ii)
There are **two / three** types of specific latent heat

[1]

(iii)
The specific latent heat of fusion is present during the **melting / condensing** of a substance

[1]

(iv)
The specific latent heat of vapourisation is present during the **freezing / boiling** of a substance.

[1]

[4 marks]

Question 1d

The student has three substances, aluminium, copper and water set up for an experiment in his laboratory where the room temperature is initially 20 °C. 500 J of heat energy is applied to each substance.

The temperature of aluminium increases to 30 °C, the temperature of the copper to 25 °C and the water to 100 °C.

Write the numbers 1, 2 and 3 next to the correct substances with 1 being the lowest and 3 being the higher specific heat capacity.

- Aluminium ....
- Copper ....
- Water ....

[3 marks]

Question 2a

The diagram in Fig 1.1 shows the changes of state between solids, liquids and gases.

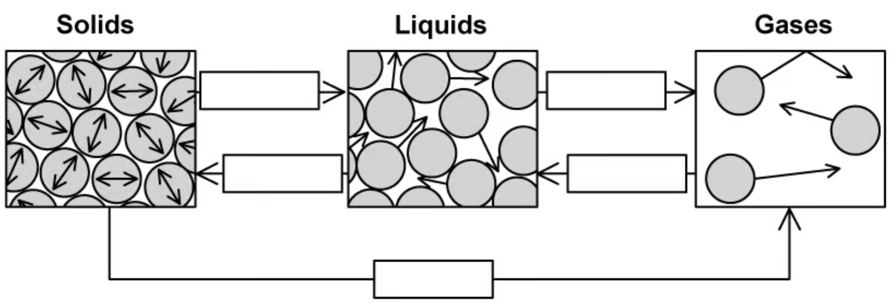


Fig. 1.1

Complete the gaps in Fig 1.1 with the names of these changes of state.

[5 marks]

Question 2b

Fig. 1.2 shows a graph of temperature against heat supplied for a substance as it goes through phase changes.

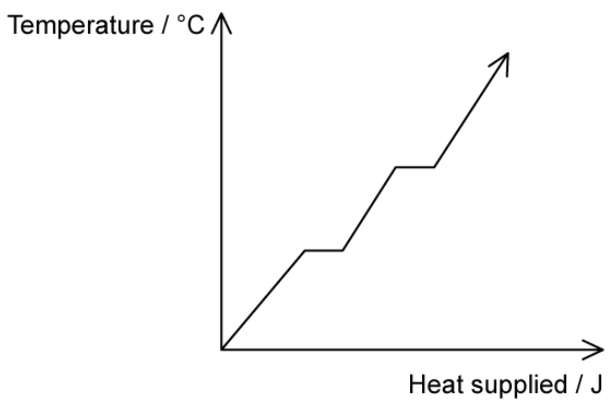


Fig 1.2

Identify, by placing the letters in the correct location on the diagram, the parts of the graph that indicates:

(i)
Latent heat of fusion. Label this A. [1]

(ii)
Latent heat of vapourisation. Label this B. [1]

[2 marks]

Question 2c

Define the specific latent heat of vapourisation.

[2 marks]

Question 2d

Define the specific latent heat of fusion.

[2 marks]

Question 3a

Different substances have different values of specific heat capacity.

Identify, by placing a tick (✓) in Table 1.1, next to the statements that correctly describe the specific heat capacity of different substances.

Table 1.1

Statements	(✓) here if the statement is correct
If a substance has a low specific heat capacity it heats up and cools down slowly.	
If a substance has a high specific heat capacity, it heats up and cools down slowly.	
The specific heat capacity of different substances determines how useful they would be for a specific purpose.	

[2 marks]

Question 3b

Use your answers to part (a) to complete this question.

(i)
One of the statements in incorrect.

Determine the correct statement by writing it in the space below.

[1]

(ii)
State a situation when it can help to know about the specific heat capacities of different materials.

[1]

[2 marks]

Question 3c

A beaker containing 0.5 kg of liquid is heated with 3000 J of energy through a temperature change of 75 °C.

Calculate the specific heat capacity of the liquid.

specific heat capacity = ... J kg^{-1} $°C^{-1}$

[4 marks]

Question 3d

A different liquid of mass 0.75 kg requires 60 000 J to boil.

Calculate the specific latent heat of vapourisation of the liquid.

specific latent heat of vapourisation = ... J kg^{-1}

[4 marks]

Model Answers: Easy

1a

(a) A table that would score 4 marks should look like:

	Equation	Definition
Specific Latent Heat Capacity	$Q = Lm$ [1 mark]	The thermal energy required to change the state of 1 kg of mass of a substance without any change of temperature. [1 mark]
Specific Heat Capacity	$\Delta Q = mc\Delta\theta$ [1 mark]	The amount of thermal energy required to raise the temperature of 1 kg of a substance by 1 °C. [1 mark]

- One mark for each piece of information in the correct place

[Total: 4 marks]

No marks are awarded for the correct definition in the equation column and vice versa. Your answers **must** be in the correct column.

1b

(b) A table that that would score 2 marks should look like:

Statement	(✓) here if the statement is correct

All substances have the same specific heat capacity.	
The heavier the material, the more thermal energy is required to raise its temperature.	✓ [1 mark]
The larger the change in temperature, the higher the thermal energy will be required to achieve this change.	✓ [1 mark]
Specific heat capacity is mainly used in gases.	

- One mark awarded for each correct statement ticked

[Total: 2 marks]

All substances have **different** specific heat capacities and specific heat capacity is mainly used in **liquids and solids**.

1c

(c) The explanations that correctly describe latent heat capacity are as follows:

(i) When a substance changes state there is **no temperature change**; [1 mark]

(ii) There are **two** types of specific latent heat; [1 mark]

(iii) The specific latent heat of fusion is present during the **melting** of a substance; [1 mark]

(iv) The specific latent heat of vapourisation is present during the **boiling** of a substance; [1 mark]

[Total: 4 marks]

1d

(d) Numbers next to the substances to identify the correct order would look like this:

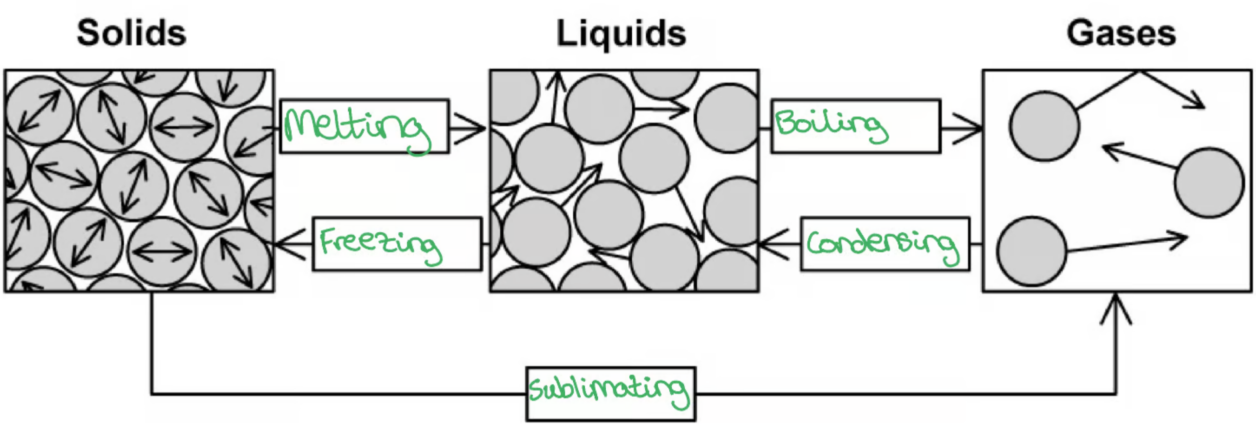
- Aluminium: **2** [1 mark]
- Copper: **1** [1 mark]
- Water: **3** [1 mark]

One mark awarded for each substance in the correct position

[Total: 3 marks]

2a

(a) A completed diagram with the changes of state added in the correct places that would score 5 marks should look like this:

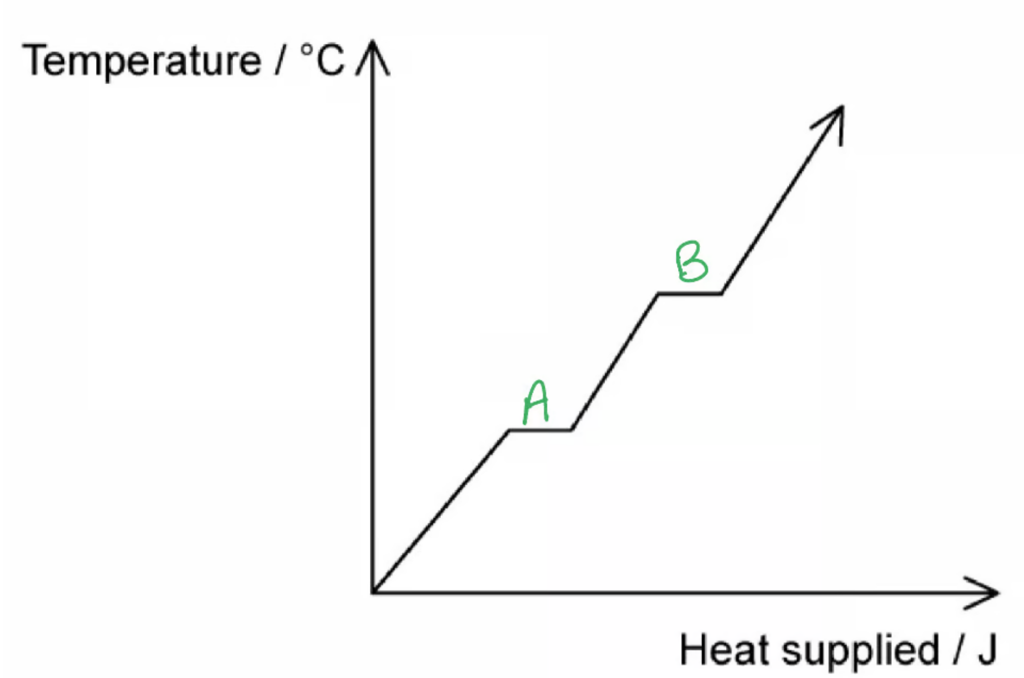


- One mark for each change of state added in the correct place

[Total: 5 marks]

2b

(b) A graph that would score 2 marks should look like this:



- A labelled on the first horizontal line from the start of the graph; [1 mark]
- B labelled on the second horizontal line from the start of the graph; [1 mark]

[Total: 2 marks]

2c

(c) The specific latent heat of vapourisation is defined as:

- The thermal energy required to convert 1 kg of liquid; [1 mark]
- To gas with no change in temperature; [1 mark]

[Total: 2 marks]

This is also the thermal energy required to convert 1 kg of **gas** to **liquid** with no

(c) The specific latent heat of vapourisation is defined as:

- The thermal energy required to convert 1 kg of liquid; [1 mark]
- To gas with no change in temperature; [1 mark]

[Total: 2 marks]

This is also the thermal energy required to convert 1 kg of **gas** to **liquid** with no temperature as well. Either definition will get you full marks.

2d

(d) The specific latent heat of fusion is defined as:

- The thermal energy required to convert 1 kg of solid; [1 mark]
- To liquid with no change in temperature; [1 mark]

[Total: 2 marks]

This is also the thermal energy required to convert 1 kg of **liquid** to **solid** with no temperature as well. Either definition will get you full marks.

3a

(a) A table that would score 2 marks should look like:

Statements	(✓) here if the statement is correct
If a substance has a low specific heat capacity it heats up and cools down slowly.	
If a substance has a high specific heat capacity, it heats up and cools down slowly.	✓ [1 mark]
[illegible]	

The specific heat capacity of different substances determines how useful they would be for a specific purpose.	✓ **[1 mark]**

- *One mark for each correct tick*

[Total: 2 marks]

3b

(b)

(i) The correct statement is:

- If a substance has a low specific heat capacity, it heats up and cools down **quickly**; [1 mark]

(ii) It can help to know about the specific heat capacities of different materials when:

Any **one** from:

- Choosing the best material for appliances / tools / construction; [1 mark]
- Determining the material composition of a substance; [1 mark]
- Making calculations related to heating / cooling a substance; [1 mark]

Any suitable answer is permitted.

[Total: 2 marks]

3c

(c) Calculate the specific heat capacity of the liquid:

List the known quantities:

- Mass of liquid, $m = 0.5$ kg
- Thermal energy supplied, $\Delta Q = 3000$ J
- Temperature change, $\Delta\theta = 75$ °C

Recall the equation for specific heat capacity:

- $\Delta Q = m\,c\,\Delta\theta$ [1 mark]

Use the specific heat capacity equation:

- $c = \frac{\Delta Q}{m\,c}$ [1 mark]
- $c = \frac{3000}{0.5 \times 75}$ [1 mark]
- $c = 80\ \text{J kg}^{-1}\ °\text{C}^{-1}$ [1 mark]

[Total: 4 marks]

3d

(d) Calculate the specific latent heat of vapourisation of the liquid:

List the known quantities:

- Mass of liquid, $m = 0.75$ kg
- Energy required, $Q = 60\,000$ J

State the equation for specific latent heat:

- $Q = m\,L$ [1 mark]

Calculate the specific latent heat of fusion of the liquid:

- $L = \frac{Q}{m}$ [1 mark]
- $L = \frac{60\,000}{0.75}$ [1 mark]
- $L = 80\,000\ \text{J kg}^{-1}$ [1 mark]

[Total: 4 marks]